Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2018

BIOLOGY **9744/03**
HIGHER 2 **23 August 2018**
Paper 3 Long Structured and Free-response Questions **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Writing Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show working or if you do not use appropriate units.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4 or 5	
TOTAL	75

This document consists of **12** printed pages.

Section A

Answer **all** the questions in this section.

1 Variant Creutzfeldt-Jakob disease (vCJD) is a rare infectious disease in humans that was first described in 1996 in the United Kingdom. It is thought to be caused by the same infectious agent which causes "mad cow" disease in cattle populations, where infected cows experience progressive deterioration of the nervous system and eventual death.

The infectious agent is a misfolded form of the protein PrP, which is normally found in nerve cells. This misfolded PrP protein is known as a prion, which can be transmitted to humans when eating food contaminated with these prions. In the brain, prions may bind to normal PrP proteins and induce them to change their conformation, forming more infectious prions which propagate the disease. Prions will bind to each other to form protein aggregates which are resistant to degradation, causing the death of nerve cells.

(a) Describe how an infectious disease differs from a genetic disease.

1. **Caused by a pathogen (e.g. viruses & bacteria) vs caused by mutant alleles / mutations in the genetic material / mutagenic factors (e.g. radiation, chemical carcinogen);**
2. **Elicits an immune response vs immune response may not be elicited;**
3. **Acquired vs inherited / can be passed on to offspring;**
4. **AVP**

Max 2 [2]

(b) Discuss how prions challenge the tenets of the cell theory.

1. **Cell theory states that all living things consist of cells OR cell theory states that the smallest unit of a living thing is a cell;**
2. **However the prion challenges the cell theory in that it has living characteristic but it is not a cell / is acellular OR it has living characteristics but is a simple protein molecule;**
3. **Since they are able to replicate themselves by converting normal proteins into more prions;**

[3]

(c) Suggest how protein aggregates may be resistant to degradation in infected cells.

1. **Ref. to enzymes are unable to bind to proteins due to changed 3D conformation, hence not complementary / specific to active site;**
2. **Ref. to enzymes are unable to access the binding site of proteins due to the formation of aggregates;**
3. **Hence not recognised / accessed by enzymes which attach ubiquitin to these proteins;**
4. **Thus cannot be hydrolysed by proteasomes / proteases into short peptides / amino acids;**

Max 2

[2]

For Examiner's Use

While vCJD is described as an infectious disease, scientists have noted that there is little or no immune response in patients diagnosed with vCJD.

(d) Suggest why there is a lack of a significant immune response in these patients.

1. **Misfolded proteins are produced by host cells and are not recognised as foreign;**
2. **Misfolded proteins are found within host cells / not found on the cell surface, hence not recognised by phagocytes or antibodies;**
3. **Prions are not presented as part of MHC-peptide complexes to stimulate the adaptive immune response;**

Max 2

[2]

Similar to the prions causing vCJD, the human immunodeficiency virus (HIV) is thought to have originated from a related strain of viruses which infects wild chimpanzees.

A HIV infection is characterised by a brief initial period of viral replication. This is followed by a long period of latency, where no symptoms are manifested. Eventually, reactivation of the provirus and rapid viral replication results in immunodeficiency.

Progression of the disease in HIV-positive individuals can be monitored by measuring the concentration of HIV particles (also known as the viral load) in their blood sample. One laboratory method measures the viral load by quantifying the activity of viral reverse transcriptase using the following procedure:

1. The HIV particles are separated from other blood plasma components.
2. The viral particles are lysed by disruption of the viral envelope and the resulting lysates are added to a mixture containing a RNA template and deoxyribonucleotides.
3. Viral reverse transcriptase found in the lysate will then synthesise complementary DNA (cDNA).
4. The amount of cDNA formed is detected by adding artificially synthesised antibodies which recognise and bind to the cDNA. These antibodies are linked covalently to an enzyme, which catalyses the formation of a coloured product.
5. The colour intensity of the resulting mixture is compared to those of standard solutions, from which the concentration of reverse transcriptase in the starting mixture can be deduced.

(e) Explain how the structure of antibodies allow them to be used for the procedure described.

1. **The Fab region / variable (V) domains / antigen binding sites of the antibodies have a unique 3D conformation;**
2. **due to a specific sequence of amino acids in these regions / domains of the heavy and light chains;**
3. **This allows specific / complementary binding to cDNA of a particular nucleotide sequence;**
4. **The Fc region / constant (C) domains of the antibodies provide a site for the linking of enzymes by covalent bonds(for the detection of cDNA formed);**
5. **AVP (e.g. hinge region)**

Max 2 from pt 1 to 3; Max 3 total

[3]

Antibodies are produced naturally by B lymphocytes in the human body, after exposure to foreign antigens.

(f) Justify the claim that "B lymphocytes are exceptions to the generalisation that all nucleated cells in the body have exactly the same DNA".

1. **Mature B cells have different DNA sequences due to somatic recombination / VDJ recombination (occurring in immature B cells);**
2. **where DNA rearrangements of the V and J segments of the gene coding for the variable (V) domain of the antibody's light chain / V, D and J segments of the gene coding for variable (V) domain of the antibody's heavy chain occurs;**
3. **Activated B cells have different DNA sequences due to somatic hypermutation;**
4. **where random point mutations are introduced at a high rate in the genes coding for the variable (V) domains of both light and heavy chains;**
5. **Activated B cells have different DNA sequences due to class switching;**
6. **where recombination of the gene segments coding for the constant (C) domains of the heavy chain occurs;**

Max 4

[4]

Another method to measure the HIV viral load involves quantifying the amount of viral genetic material present in the blood sample, using the procedure in Fig. 1.1.

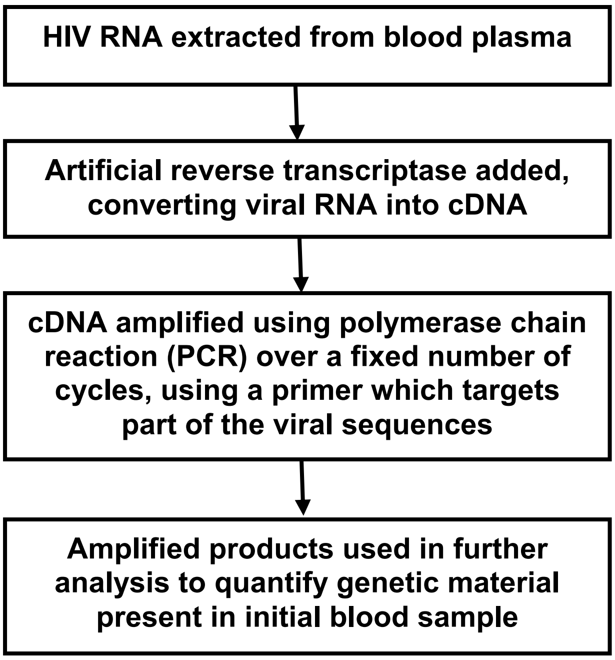

Fig. 1.1

(g) Describe how PCR is used in the amplification of HIV cDNA.

1. **HIV cDNA is incubated with *Taq* polymerase / heat-stable DNA polymerase, deoxyribonucleotides and the forward and reverse primers;**
2. **Temperature is raised (in a thermal cycler) to 95°C, which denatures the cDNA / breaks hydrogen bonds between complementary bases, to form single-stranded DNA / to separate the two strands of DNA;**
3. **Temperature is cooled to 55°C, allowing primers to anneal / form complementary base pairs with the two (3') ends of the target viral sequences;**
4. **Temperature is raised to 72°C, allowing synthesis of the daughter DNA strands by elongation of the primers;**
5. **Cycle is repeated many times, where the number of DNA molecules doubles after each cycle;**

Pt 2, 3 and 4 compulsory
Max 4

[4]

There are two major strains of HIV known to man, HIV-1 and HIV-2. The different methods used to measure viral load vary in their ability to distinguish between the different HIV strains, as shown in Table 1.1.

Table 1.1

Method of measuring viral load	Able to distinguish between HIV strains
Quantification of viral reverse transcriptase activity	No
Quantification of viral genetic material	Yes

(h) Explain the difference in the two measurement methods in their ability to detect different HIV strains.

1. **HIV strains differ in their RNA sequences / have different genes or genetic makeup;**
2. **Hence PCR primer used in the method which quantifies the viral genetic material may be designed to bind to sequences found in a particular strain but not the other; (R! mention of gene coding for RTase, integrase or protease)**
3. **In the quantification of viral reverse transcriptase activity, the reverse transcriptase of either HIV strain would be able to reverse transcribe the RNA template used in the method / Ref. to not sequence-specific / Ref. to different HIV strains have the same type of reverse transcriptase;**

Note: Reverse transcriptase is essential for viral replication, hence assumed to be functional regardless of viral strain

[3]

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While HIV is sexually-transmitted, dengue virus is transmitted by a mosquito vector. To study the influence of weather conditions on the incidence of dengue viral infection in Singapore, scientists recorded the amount of rainfall, mean weekly temperatures and the number of dengue cases over a one-year period. The recorded data is shown in Fig. 1.2.

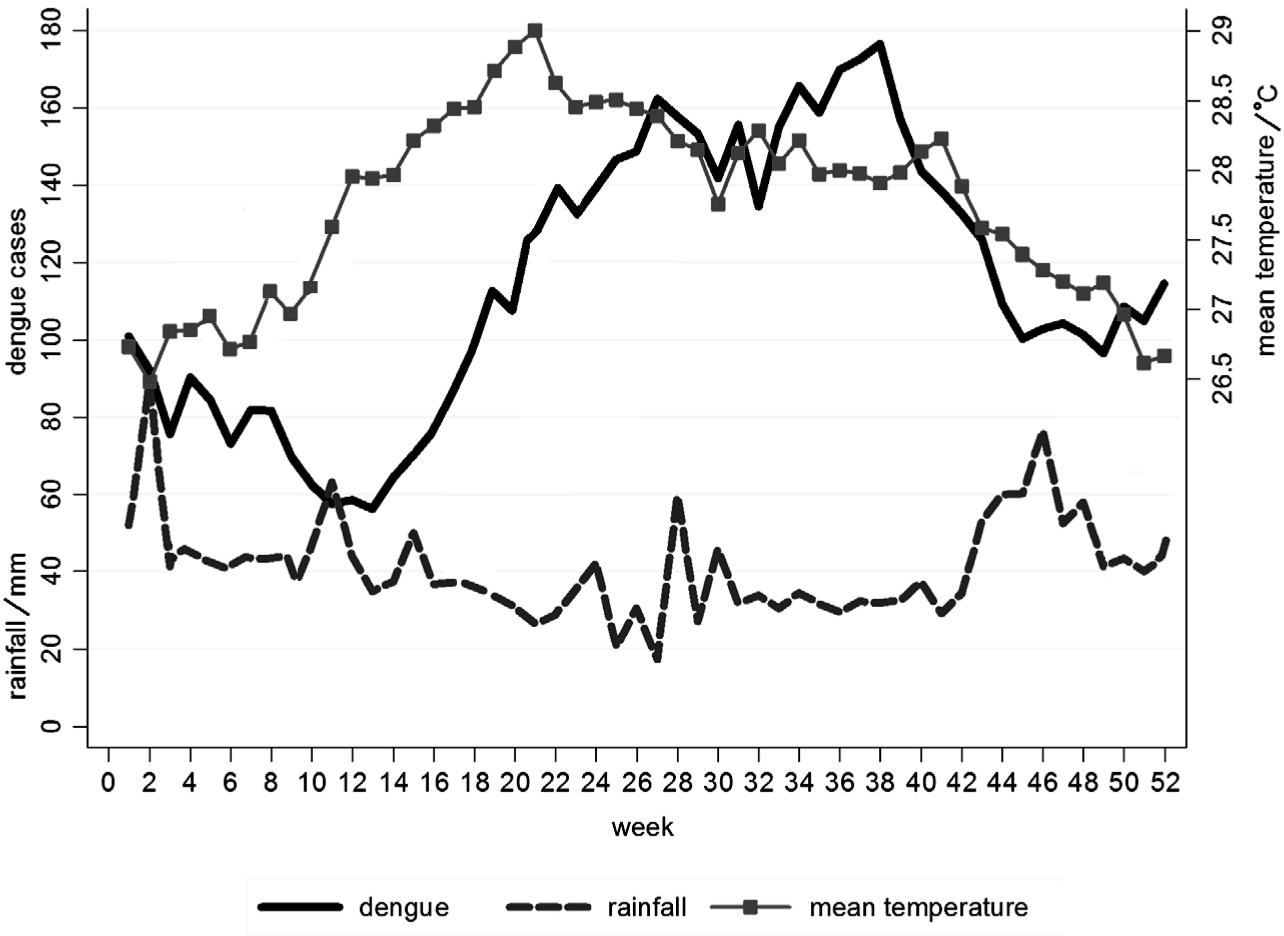

Fig. 1.2

(i) Based on the information provided, comment on which weather condition is a better predictor of dengue cases in Singapore.

1. **Mean weekly temperatures was a better predictor of the number of dengue cases;**
2. **Quoting relevant data to support correlation between temperature and dengue cases for upward trend e.g. When temperatures increased from 26.7°C at week 1 to 28.4°C at week 27, number of dengue cases increased from 100 to 160;**
3. **Quoting relevant data to support correlation between temperature and dengue cases for downward trend e.g.When temperatures decreased from 28°C at week 40 to 27.1°C at week 48, number of dengue cases decreased from 142 to 100;**
4. **Quoting relevant data to show that there is little correlation between rainfall and dengue cases (e.g. When rainfall remained relatively constant around 32mm between week 32 and week 41, number of dengue cases increased sharply from 138 to 178, before decreasing to 140);**

[4]

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(j) Suggest explanations for the relationship between the weather condition chosen in part (i) and the number of dengue cases recorded.

1. **Increase in temperature increases the rate of enzymatic activity and hence metabolism of *Aedes* mosquitos / mosquito vectors, increasing their activity and facilitating spread of dengue;**
2. **The development time of mosquito larvae shortens as temperature increases, increasing the number of mosquito vectors to spread the dengue virus;**
3. **Rate of dengue viral replication increases with increase in temperature, hence more infected individuals developing viremia for a longer period of time, facilitating spread of dengue;**

R! Increase in precipitation brought about by increased temperature leads to the creation of standing water spots for the breeding of mosquitos (not supported by data)

Max 2

[2]

[Total: 29]

2 In order to solve the problems of malnutrition and nutritional deficiencies that are faced by two billion people worldwide, scientists have turned to genetic engineering. This involves the introduction of foreign genes into organisms to increase food production. One technique used in plant biotechnology to generate transgenic crops (crops with foreign genes introduced) involves the use of protoplast culture. It is designed to work as follows:

1. Cut a leaf into pieces.
2. Soak the leaf in a solution of microbial enzymes, sugars and salts. The enzymes digest the cellulose cell wall. The sugars and salts maintain the osmotic potential.
3. Centrifuge the cells to remove the cell wall debris. The protoplasts (plant cells with cell wall removed) will be found in the supernatant.
4. Apply an electric field to introduce the foreign genes into the protoplasts via transformation.
5. Isolate transformed protoplasts and culture them into transgenic plants.

(a) **(i)** Before a foreign gene is introduced into the protoplasts, the photosynthetic activity of the protoplasts must be checked to ensure that they are viable.

One way to do this is to mix the protoplast extract with 2,6-dichlorophenolindophenol (DCPIP), a redox dye. When reduced, DCPIP is changed from blue to colourless.

A protoplast mixture with DCPIP is placed under unfiltered white light, while another mixture is placed under white light passed through a blue filter.

Assuming that light intensity of the white light is constant, explain whether the DCPIP in the protoplast extract would decolourise faster under the unfiltered or filtered white light.

1. **Unfiltered white light;**
2. **Unfiltered white light has a broader spectrum which includes red and blue light;**
3. **The rate of light-dependent reactions / photophosphorylation and photolysis is increased;**
4. **The rate of electron emission from chlorophyll a would be faster and this will allow for a faster rate of DCPIP being reduced and decolourised;**

[4]

(ii) Suggest why it is necessary to digest the cellulose cell wall to allow for the introduction of a foreign gene into the protoplast.

1. **DNA molecule is large and may not pass through the cell wall easily / effectively; (R! cell wall is impermeable)**
2. **To expose the selectively permeable membrane;**
3. **To increase the efficiency of the transformation process / OWTTE;**

Max 1 [1]

Using the protoplast culture technique, scientists have introduced three β-carotene biosynthesis genes into a particular variety of the rice plant to form a new variety – the golden rice plant. The golden rice plant looks morphologically similar to other rice varieties except that its rice grains are yellow in colour due to the production of β-carotene in its rice grains. The β-carotene contained in the rice grains will be converted into vitamin A when it is consumed by humans.

(b) Discuss whether the β-carotene biosynthesis genes will be preserved in the gene pool of a population of rice plants, containing both the original and the new varieties, without any human intervention.

Yes, will be preserved

1. **The genes do not confer any advantageous traits nor disadvantageous traits on the growth and reproductive success of the rice plant. Hence, the trait does not come under selection pressure;**
2. **The golden rice plant also looks morphologically similar to the other rice variety, hence animals will feed equally on both varieties of plants. Hence, the trait does not come under selection pressure;**
3. **Animals that feed on rice preferentially consume the most common varieties of plants. Golden rice is rare and thus tend to be at a selective advantage;**
4. **AVP; (Must have idea of selected for + reason why)**

Max 2m

No, will not be preserved

5. **Animals may preferentially consume the golden rice plant (due to valid reason given). Hence, it be selected against in the natural environment;**
6. **The plant may spend available resources in producing β-carotene that it does not require. Hence, it be selected against in the natural environment;**
7. **Genetic drift may occur and a chance event may eliminate the entire Golden rice population which is small;**
8. **AVP; (Must have idea of selected against + reason why)**

Max 2m [4]

Genetic engineering can also be used to treat genetic diseases in humans which are caused by mutations involving single genes. This is done by introducing copies of the normal allele into human cells. In germline cell therapy, the normal alleles are introduced into germ cells.

(c) Comment on the ethical aspects of germline cell therapy.

It is ethically unacceptable:

1. **because there is a possibility of introducing an inheritable mutation/trait and all descendants of the patient would be affected;**
2. **due to unknown risks of altering the genome;**
3. **as it may be used for genetic enhancement / enhancement of traits in the offspring by altering the genome;**
4. **due to potential exploitation of women for collection of their oocytes;**

R! desensitise the society to more unethical practices
R! harming of embryos

It is ethically acceptable:

5. **because it may be used to treat incurable conditions/ leads to savings in healthcare cost;**
6. **AVP;**

Max 2m [2]

[Total: 11]

3 Plants have long been regarded as carbon sinks because they take in carbon dioxide for photosynthesis. However, when temperatures rise, plants increase their rate of respiration, resulting in increased carbon release. Some research has suggested that this could convert forests from a long-term carbon sink to a carbon source, aggravating climate change.

(a) Outline how a rise in temperature could lead to increased carbon release by plants.

1. **An increase in temperature leads to increased kinetic energy and hence an increase in effective collision between enzyme and substrate/ rate of enzyme-substrate complex formation;**
2. **This leads to an increase in rate of the link reaction/ Krebs cycle;**
3. **More carbon is released in the form of carbon dioxide from oxidative decarboxylation reactions;**

[3]

In 2016, a team of scientists conducted a short-term study of five years to find out the net carbon exchange of trees when the temperature was increased. In order to determine this, the increase in leaf respiration at higher temperatures was evaluated using 1000 young trees of 20 different boreal and temperate tree species grown in an open-setting.

Scientists compared the data observed to the expected results in Fig. 3.1.

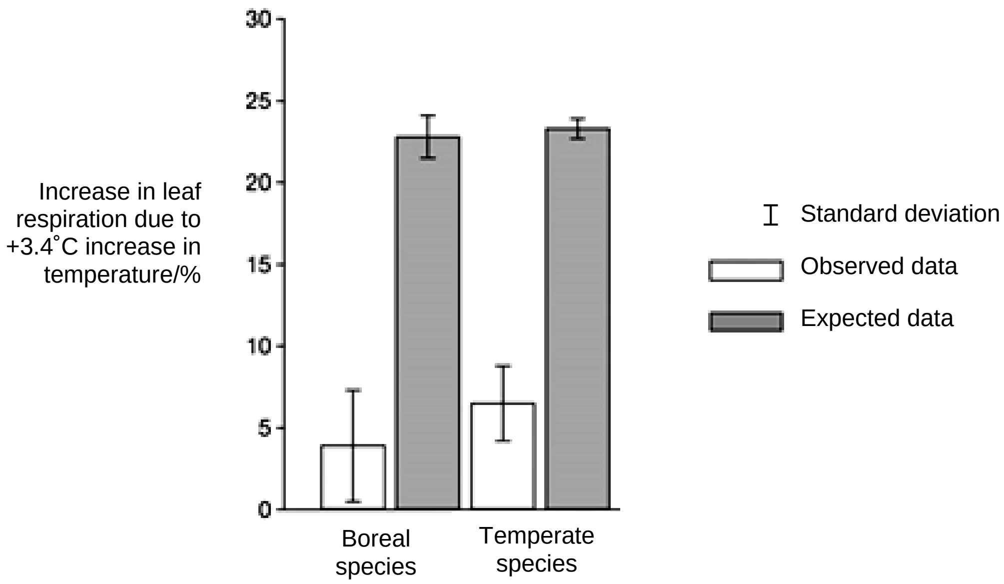

Fig. 3.1

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(b) (i) With reference to Fig. 3.1, describe one difference between the data observed and the expected results.

1. **For both types of trees, the expected increase in leaf respiration was higher than that of the data observed;**
2. **The expected increase in leaf respiration was 23 and (22.5 to) 24% while that observed showed an increase in (3.5 to) 4% and (6.5 to) 7% respectively for boreal and temperate species;**

OR

3. **For both types of trees, the standard deviation for expected results was smaller than that for the data observed;**
4. **The standard deviation for expected data was (3 to) 4% and (1 to) 2% respectively for boreal and temperate species compared to 7 (to 8%) and 5% for the data collected;**

Pt 1 & 2 OR Pt 3 & 4

[2]

(ii) In Fig. 3.1, the data observed shows a difference in the increase in leaf respiration between boreal and temperate tree species. Explain whether this difference is significant.

1. **The difference is not significant as the standard deviation bars overlaps;**

[1]

(iii) Based on the results of the study, comment on whether forests will remain as carbon sinks or be converted to carbon sources if temperatures rise.

1. **The rate of plant respiration did not increase as much as expected, suggesting that the rate of photosynthesis may still be higher than the rate of respiration;**
2. **Overall, plants will take in more carbon dioxide than it gives out / remain as carbon sinks;**

OR

3. **The rate of plant respiration did not increase as much as expected. However, there is still an increase in respiration which may result in the rate of respiration becoming higher than the rate of photosynthesis;**
4. **Plants might become carbon sources instead of carbon sinks;**

Pt 1 & 2 OR Pt 3 & 4

[2]

(iv) Based on the results, scientists concluded that plants can acclimatise to changing climates, adjusting their metabolism according the environment. It was thus suggested that crop yield and hence food supply would not change drastically in light of climate change.

With reference to the design of the experiment, discuss the validity of this conclusion.

<u>Valid</u>

1. **A large sample size of 1000 trees were used, increasing the confidence of this conclusion;**
2. **AVP;** **Max 1m**

<u>Invalid</u>

3. **This study was only conducted on 20 boreal and temperate species and a conclusion cannot be made for plants in general;**
4. **This was a short term study and there is no evidence on the long term implication;**
5. **The study was conducted in an open setting, which might be affected by changes in other variables besides temperature;**
6. **AVP;** **Max 1m**

[2]

Section B

Answer **one** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

4 Biological polymers are large molecules composed of many individual smaller molecules called monomers. For instance, amino acids are taken into an animal cell for the synthesis of collagen. The synthesised collagen is then secreted out of the cell to form the connective tissue.

(a) Explain how the structural features of amino acids contribute to the function of collagen. [10]

(b) Explain how temperature and pH may affect the transport of substances across the cell surface membrane of eukaryotes. [15]

[Total: 25]

5 There are similarities in the ways that prokaryotes and eukaryotes regulate gene expression. In both groups of organisms, cyclic adenosine monophosphate (cAMP) is an important molecule in many biological processes, including the regulation of gene expression.

(a) Describe the role of cAMP in altering gene expression in organisms and suggest why they are usually small and non-protein in nature. [10]

(b) The eukaryotic genome is considered to be more complex than the prokaryotic genome. Describe the differences between the eukaryotic and prokaryotic genomes and explain the significance of these differences. [15]

[Total: 25]

4(a) Explain how the structural features of amino acids contribute to the function of collagen. [10]

1. Collagen has a structural role;

Structure of amino acid	Features of amino acids	Properties of collagen
2. Amino group and carboxyl group **(need to link point 2 with other points)**	**3. Amino and carboxyl groups allow for peptide bond formation between them;** **4. Via a condensation reaction;** **5. Allowing for the joining of (~1000) amino acid residues to form an alpha-chain;** **6. –NH and –CO groups are involved in forming hydrogen bonds between alpha-chains;**	**7. Allows formation of large molecule;** **8. This renders collagen insoluble in water;** **9. Three α-chains can wind tightly together to stabilise the triple helix structure (A! ref. to being compact);** **10. These properties account for the high tensile strength in collagen;**
11. R group (which may be polar or non-polar)	**12. The α-chains of collagen have a tripeptide sequence of gly-X-Y, where X is often proline and Y is often hydroxyproline or hydroxylysine;** **13. Glycine and proline have non-polar R groups that render them hydrophobic;** **14. Polar R groups (such as hydroxyproline and hydroxylysine) are already involved in hydrogen bonding between α-chains;** **15. Glycine has a small R group that is found in every third position;** **16. R groups of amino acid residues (lysine) allow for formation of covalent cross-links between tropocollagen;**	**17. This renders collagen insoluble in water;** **18. Three α-chains can wind tightly together to stabilise the triple helix structure;** **19. These properties account for the high tensile strength in collagen;**

*** Properties to be awarded once (pt 8, 9 and 10 vs pt 17, 18 and 19)**

+QWC: Mentions at least 1 structural features of amino acids and linking correctly to one property of collagen (insolubility in water and high tensile strength)

(b) Explain how temperature and pH may affect the transport of substances across the cell surface membrane of eukaryotes. [15]

Transport that does not involve proteins	Transport that involves proteins
1. Diffusion is the net movement of particles (molecules or ions) from a region where they are at a relatively high concentration to a region where they are at a lower concentration; **2. Osmosis is the movement of water from a region of higher water potential (Ψ) to a region of lower water potential;** **3. When temperature increases, kinetic energy of particles increases;** **4. Rate of diffusion of particles (for diffusion) and water molecules (for osmosis) increases across cell surface membrane;** **5. Endocytosis (including phagocytosis and pinocytosis) is the uptake of mostly large material into a cell by an invagination of the plasma membrane / engulfing and its internalisation in a membrane-bound vesicle;** **6. Exocytosis is the process where the cell secretes certain large molecules by the fusion of vesicles with the cell surface membrane;** **7. When temperature increases, fluidity of membrane increases because phospholipid movement increases;** **8. Hence increasing the rate of endocytosis and exocytosis of substances;** **9. If the temperature is high enough, hydrophobic interactions between phospholipid tails may be disrupted;** **10. This causes membrane leakages, allowing larger substances to enter or leave the cell;**	**11. Facilitated diffusion of substances occurs with the assistance of channel / carrier proteins in the membrane, along a concentration gradient;** **12. Active transport is the transport of substances into and out of cells regardless of the presence of a concentration gradient with the expenditure of ATP;** **13. In receptor-mediated endocytosis, specific molecules/ligands bind to specific receptors / glycoproteins / glycolipids before being taken up by the cell;** **14. At high temperatures; the hydrogen bonds, hydrophobic interactions and ionic bonds (name 2 out of 3) between amino acid residues of proteins are broken;** **15. When there are drastic changes in pH, ionic charges of the acidic and basic R-groups of the amino acid residues at the binding site may be changed;** **16. Changes in pH also affect hydrogen bonds and ionic bonds between amino acid residues (name 1 out of 2);** **17. These changes in temperature / pH can denature / change the 3D conformation of the channel and carrier proteins involved in facilitated diffusion or active transport;** **18. E.g. of transport protein affected;** **19. Hence increasing / decreasing the rate of facilitated diffusion / active transport;**

Points 3 and 4 apply for both diffusion and osmosis; Accept reverse argument for all points

+ QWC: Mentions at least 3 different modes of transport that allows explanation of effects of temperature and pH

5(a) Describe the role of cAMP in altering gene expression in organisms and suggest why they are usually small and non-protein in nature. [10]

1. **In eukaryotes, it acts as a second messenger;**
2. **In the presence of the ligand (first messenger), an activated G protein binds to adenyl cyclase, converting ATP to cAMP;**
3. **cAMP binds to and activates protein kinase A during signal transduction;**
4. **Ref. to idea of signal amplification + example (e.g. one adenyl cyclase catalyses the synthesis of many molecules of cAMP / one protein kinase phosphorylates more than one kinase and activates them);**
5. **Protein kinase A then activates other protein kinases via a phosphorylation cascade;**
6. **Finally activating an effector / activator / repressor / transcription factor protein that alters gene expression;**
7. **In the absence of the ligand, cAMP will be converted to an inactive product (AMP) by phosphodiesterase to cease signal transduction;**

8. **In bacteria, it is involved in the positive regulation of the *lac* operon;**
9. **When glucose levels are low (and lactose is present), ATP is converted to cAMP by adenyl cyclase;**
10. **cAMP binds to and activates the catabolite activator protein (CAP) which binds to the CAP binding site within the *lac* promoter;**
11. **This increases the affinity of RNA polymerase to bind to the *lac* promoter and increases rate of transcription of structural genes;**

12. **cAMP is small in nature as they need to diffuse rapidly through the cytoplasm, enabling quick transduction of signal throughout the cell;**
13. **cAMP is non-protein in nature as proteins are sensitive to conditions in the cell (such as pH and temperature), hence activity may be easily affected/proteins are easily degraded;**
14. **cAMP is small in nature as larger molecules may take a longer time to synthesise and hence unsuitable for rapid action;**
15. **cAMP is non-protein in nature as protein synthesis may take a longer time and hence unsuitable for rapid action;**
16. **cAMP is small in nature as larger molecules requires a lot of energy/resources to synthesise;**
17. **cAMP is non-protein in nature as protein synthesis requires a lot of energy/resources;**

18. **Ref to Glucagon binding to GPLR and its role in regulating blood glucose concentration;**

Max 4 for eukaryotes
Max 3 for small, non-protein nature
Max 9 in total

+ QWC:
1 role of cAMP in eukaryotes and 1 role of cAMP in prokaryotes (bacteria)
1 reason why it is small or why it is non-protein in nature

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(b) The eukaryotic genome is considered to be more complex than the prokaryotic genome. Describe the differences between the eukaryotic and prokaryotic genomes and explain the significance of their differences. [15]

Level of comparison	Prokaryotic Genome	Eukaryotic Genome	Significance
1. Size of genome	**Smaller**	**Larger**	**1a. In eukaryotes, there are more genes / coding regions;** **1b. Allows more proteins need to be coded for;**
2. Number of chromosomes	**Single / haploid**	**Multiple / diploid or more**	**2a. In eukaryotes, there is pairing of homologous chromosomes;** **2b. Allows for crossing over of non-sister chromatids during meiosis, giving rise to genetic diversity;** **2c. In prokaryotes, there is a smaller genome which allows for faster binary fission;**
3. Organisation of genes/operon	**Polycistronic mRNA / operons present**	**Monocistronic mRNA / operons absent**	**3a. In prokaryotes, there is coordinated control of metabolic enzymes involved in the same pathway;** **OR:** **In eukaryotes; individual enzymes are controlled individually by separate promoters (which can be produced in different amounts or not at all);** **3b. In prokaryotes, a coordinated control allows for a more rapid response as all substrate will be channeled into that one single pathway;** **3c. In eukaryotes, this way of organization allows for different metabolic intermediates to be used in different metabolic pathways to suit the cells' needs;**
4. Amount of coding and non-coding DNA	**Mainly coding, little non-coding DNA;**	**Greater proportion of non-coding than coding DNA;**	**4a. In eukaryotes, non-coding DNA in eukaryotes with regulatory functions, such as enhancers / silencers allows for different levels of expression of a particular gene depending on the activators / repressors present in the cell for a specific cell response;** **4b. In eukaryotes, the presence of introns allows for alternative splicing to occur, allowing one**

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			gene to code for more than one type of mature mRNA / protein; 4c. In eukaryotes, specific proteins bind to centromere and forms the kinetochore which is the site of spindle fibre attachment for the accurate segregation of sister chromatids during mitosis and chromatids in meiosis II / the segregation of homologous chromosomes during meiosis I to opposite poles, and hence to daughter nuclei; OR Centromere also allow the adhesion of sister chromatids to ensure accurate segregation of sister chromatids during mitosis and chromatids in meiosis II 4d. In eukaryotes, centromeres also help organise the chromatin within the nucleus during interphase; 4e. In eukaryotes, telomeres protect genes at the ends of chromosomes from being eroded / other functions 4f. But prokaryotes have circular DNA and do not face the end-replication problem;
5. Association of genetic material with histones	**No association with histones**	**Association with histones**	**5a. In eukaryotes, this allows for folding of DNA to higher degree of condensation so that there is less breakage during cell and nuclear divisions;** **5b. Control of rate of gene transcription by allowing for increased DNA condensation/conversion between euchromatin and heterochromatin states;**
6. Location of genetic material	**In nucleoid region**	**In (membrane-bound) nucleus**	**6a. Without the nuclear envelope, transcription and translation takes place simultaneously;** **6b. Translational control is less significant in prokaryotes;**

R! Comparisons without reference to the genome, such as size of cell

+QWC: Explain at least one difference between eukaryotic and prokaryotic genomes with corresponding significance(s)